

Quantitative Cytoarchitectonics of the Posterior Cingulate Cortex in Primates

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ABSTRACT

The cytoarchitecture of four cortical areas within the posterior cingulate region and the differences among them are quantitatively analyzed in 17 primate species. The transition from allocortex (area 29) to isocortex (areas 30, 23, and 31) is characterized by shifts in laminar proportions and cellular densities. The molecular layer and inner main lamina (layers V and VI) occupy a smaller proportion of the cortex in iso- than in allocortex, while conversely, the outer main lamina (layers II and III) makes up a larger proportion of the cortex in iso- than in allocortex. Layer IV is relatively larger in area 31 than in area 23. The packing densities of cell bodies were determined by gray-level index values. These values show that the isocortex has lower cell densities in the outer main lamina, but higher cell densities in the inner main layers. Furthermore, the granular layer of area 31 is more densely packed than it is in area 23. These results are discussed as structural correlates of a relative increase in the receptive and processing capacities of the isocortex over the adjacent allocortex.

Key words: limbic cortex, cytoarchitecture, quantitative morphology

Quantitative morphological studies of structures within Papex's circuit have shown that several of its subcortical and cortical regions are larger within the human brain than expected for a primate with a brain of this size (Hopf et al., '67; Stephan, '75; Passingham, '79; Armstrong, '80, '82, '86). To extend these comparisons to further limbic cortical regions, particularly the posterior cingulate cortex, it is necessary to define the structure of these regions quantitatively. Qualitatively systematic descriptions of the posterior cingulate cortex have already been made in many mammals including primates (Brodmann, '05, '08, '09; Rose, '27; Rose and Woolsey, '48; Sanides, '69; Stephan, '75; Vogt, '76; Braak, '80). This paper extends these observations by presenting morphometric data about this region.

Recent advances in image analysis (Zilles et al., '84) allow a quantitative investigation of these regions. Common aspects of construction found in these areas and in the transition from allo- to isocortex were analyzed in 17 different primate species. These results are presented here and form the basis for a quantitative comparison of each area among these species in the following paper (Armstrong et al., '86).

The posterior cingulate region of primates has both allocortical (e.g., Brodmann's area 29) and isocortical areas (areas 30, 23, 31) and therefore shows different forms of

cytoarchitecture. Specifically area 29 (granular retrosplenial cortex) is classified as periarchicortex, area 30 (agranular retrosplenial cortex) as proisocortex, and areas 23 and 31 as isocortex (Stephan, '75). Within the callosal sulcus area 29 is deepest or most lateral and, on its medial boundary, abuts area 30. Area 30 is also within the callosal sulcus and is bordered medially by area 23. In most species, area 23 forms the medial surface of the cingulate gyrus and is bordered dorsally by area 31.

MATERIALS AND METHODS

Seventeen different primate species were studied. They include the prosimians *Lemur catta*, *Loris tardigradus*, *Perodicticus potto* ($n=2$), *Nycticebus coucang*, and *Tarsius syrichta*, and the following anthropoids: *Callithrix jacchus* ($n=2$), *Aotus trivirgatus*, *Alouatta seniculus*, *Saimiri sciureus*, *Cebus albifrons*, *Cercopithecus mitis* ($n=3$), *Erythrocebus patas*, *Macaca mulatta* ($n=2$), *Cercocebus albigena*, *Papio species* ($n=2$), *Pan troglodytes*, and *Homo sapiens*. The anthropoids have been divided into their infraorders:

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platyrrhine (New World monkeys, $n=5$) and catarrhine [Old World monkeys ($n=5$); apes ($n=1$); and human ($n=1$)]. All the brains were serially sectioned at 20 μm and stained with cresyl violet.

The areas 29, 30, 23, and 31 (Brodmann, '05, '08, '09) of the posterior cingulate cortex were identified and analyzed. The detailed analysis of each brain was carried out on at least two sections taken immediately rostral to the splenium of the corpus callosum. The volumetric proportions of cortical layers and gray-level indices (GLI) (Schleicher et al., '78; Zilles et al., '78, '79a, '79b, '82) were measured on these sections. They were digitized by a computer-controlled scanning stage and a television image analyzer (Microvideomat 2, C. Zeiss, Oberkochen, F.R.G.).

At each position of the scanning stage, the GLI was measured in a field 20 $\mu\text{m} \times 20 \mu\text{m}$. The GLI expresses the proportion of the measuring field occupied by profiles stained with cresyl violet; neuronal cell bodies and glial and endothelial nuclei. Following this, the GLIs were stored in a picture matrix and printed. The resulting computer plots show the regional distribution of the GLIs classified into five arbitrary groups on the basis of increasing mean GLIs. These classifications are represented in the computer printouts by increasing densities. Following this classification, the boundaries of the different layers and areas were delineated by superimposing the computer plots upon the image of the histological slide by using a drawing tube attached to a microscope. In each section the full extent of each lamina was studied. Layers in different areas selected for further analysis were then outlined with a cursor.

Within the defined regions, areal and mean GLI values were calculated by this method. After the areas of the cortical laminae had been determined, the volumetric proportion of each layer to the cortical field was determined according to the equation

$$V(p) = [V(1) \times T \times 100] / V(f) \times T \quad (1)$$

where $V(p)$ is the volumetric proportion of a cortical layer in this field, $V(1)$ is the volume of layer 1 in this field, and $V(f)$ is the total volume of the cortex in the sections. Since T is constant the volumetric proportion $V(p)$ equals the areal proportions $A(p)$. In the following both expressions are used.

For the present analysis we defined the following layers in the cortical areas: molecular, granular, and outer and inner main laminae. The molecular layer is layer I. In area 29 the outer main lamina corresponds to Vogt's ('76) layer 2 (III [IV]) and in areas 30, 23, and 31 it corresponds to layers II + IIIa-c of Vogt's ('76) description in the rhesus monkey. The inner main lamina corresponds to layers V and VI (3 + 4 in area 29).

In addition to helping generate cytoarchitectural divisions, the GLI values were also transformed into standardized GLI values. That is, in each cortical area of each species the GLI values of the different layers were measured and related to the mean weighted GLI value of the whole cortical area 29 according to the equation

$$GLI(s) = [GLI(1) \times 100] / GLI(f) \quad (2)$$

where $GLI(s)$ is the standardized GLI of the cortical layer in percent, $GLI(1)$ is the mean GLI of layer 1 in percent, and $GLI(f)$ is the mean GLI of the whole area 29 in percent.

In this paper, the term GLI value is synonymous with standardized GLI value.

To compare the differing proportions of area and GLI values among the four cortical areas, a trend analysis was carried out in each species according to the following procedure:

$$x = (-3t_{129} - t_{130} + t_{123} + 3t_{131}) - (-3t_{m29} - t_{m30} + t_{m23} + 3t_{m31}) \quad (3)$$

where t is either the volumetric proportion or GLI of the cortical layer in the identified cortical areas (e.g., t_{129} = value for layer 1 in area 29, and t_{m29} is the volumetric proportion or GLI of cortical layer m in area 29). From the trend analysis, we determined whether the outer and inner main laminae are similarly proportioned in all cortical areas. Likewise, we also tested whether the molecular layer occupies the same proportion of the outer plus inner main laminae. If the proportions do not change, then the slopes produced by the trend analysis of two layers are the same and the mean difference ($\bar{d}$) equals 0. The significance of mean differences in the cortical areas under study were determined by the following equation:

$$\frac{\bar{d}}{s/\sqrt{n}} \sim t_{n-1} \text{ (d.f.)} \quad (4)$$

where $\bar{d}$ is the difference between the slopes, s is the standard error of the distribution, and n is the number of proportions.

A granular layer is only present in two of the areas under investigation (areas 23 and 31). Consequently, the significance of the relative size differences in areal proportions and GLI values between these two areas was tested with the Wilcoxon signed-rank test.

RESULTS

Gyral development

The observed primate brains differ according to the degree of gyral depth on the medial surface of the posterior cingulate region. In the following "lissencephalic" means either a completely smooth cortical surface or a surface with a small dimple. A gyrencephalic surface minimally has a clearly visible cingulate sulcus.

In prosimians this medial surface is generally lissencephalic. *Tarsius syrichta*, *Lemur catta*, *Loris tardigradus*, and *Nycticebus coucang* have, at most, a small dimple. *Perodicticus potto* has a variable development of the cingulate sulcus; one individual was observed to have a cingulate dimple while the other had a small sulcus.

The small platyrrhines (New World monkeys), *Callithrix jacchus*, *Aotus trivirgatus*, and *Saimiri sciureus*, are lissencephalic, while the larger-brained platyrrhines, *Alouatta seniculus* and *Cebus albifrons*, have cingulate sulci. All members of the Cercopithecoidea (Old World monkeys) and Hominoidea (apes and humans) are gyrencephalic in this region.

Area descriptions and borders

In all primates, the lateral part of the corpus callosum is covered by a thin cell layer, the indusium griseum and the subiculum. Both areas are summarized in Figure 1 as supracommissural hippocampus (SH). They are bordered se-

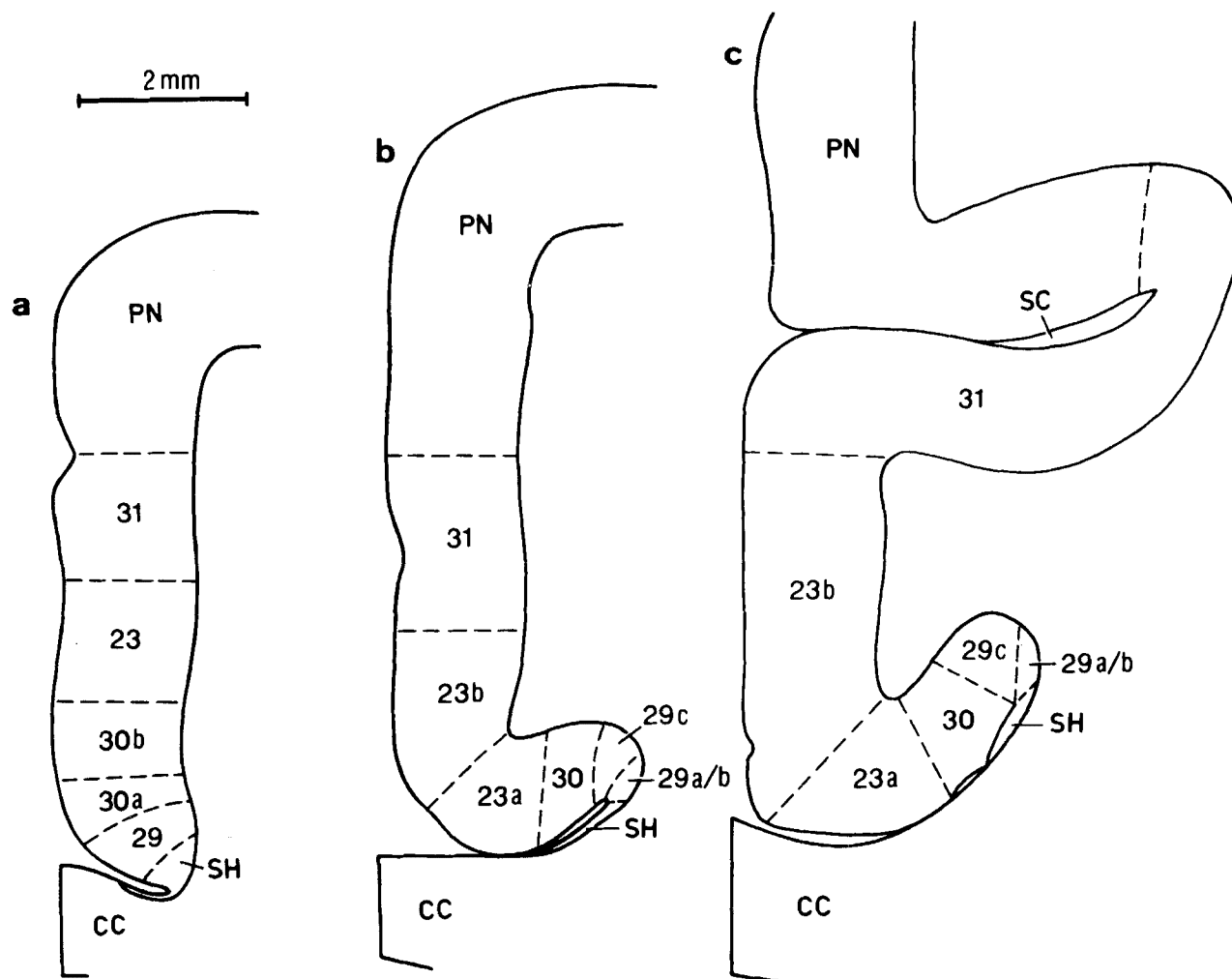


Fig. 1. Areal pattern in coronal sections of the medial cortical surface immediately rostral to the splenium of the corpus callosum. a = *Lemur catta*, b = *Saimiri sciureus*, c = *Macaca mulatta*. CC = corpus callosum,

PN = parietal neocortex, SH = supracommissural hippocampus. 29 = granular retrosplenial cortex (area 29); 30a,b = agranular retrosplenial cortex (area 30); 23a,b = isocortical area 23; 31 = isocortical area 31.

quentially by the granular retrosplenial cortex (areas 29 and 26 of Brodmann, '09), the agranular retrosplenial cortex (area 30 of Brodmann, '09), and the isocortical cortex (areas 23 and 31). The isocortical areas are next to parietal and occipital neocortex (Fig. 1). All of the granular retrosplenial cortex is termed area 29 according to Vogt ('85).

Area 29 always shows a four-layered structure in Nissl preparations (Fig. 2). Below the molecular layer is a dense external pyramidal layer with small cells, followed sequentially by the internal pyramidal layer and multiform layer. The internal pyramidal layer is also directly continuous with the pyramidal cells in the medially adjoining subiculum (Fig. 2e,i). Area 29, however, can be further subdivided on the basis of cytoarchitectural details. The lateral subarea (area 29a/b of Vogt, '85) does not clearly separate into internal pyramidal and multiform layers whereas the medial subarea does (area 29c of Vogt, '85). Additionally, a

seam of larger pyramidal cells borders the molecular layer in the external pyramidal layer of area 29c. This differentiation of the laminar structure is easily detected in anthropoids but is barely visible in most prosimians. Therefore, the following comparative study (Armstrong et al., '86) is based not on this subdivision but on area 29 as a unit.

Fig. 2. Micrographs of Nissl-stained coronal sections (20 μ m) from the posterior cingulate cortex in *Lemur catta* (a-d), *Saimiri sciureus* (e-h), and *Macaca mulatta* (i-l). Area 29 (delineated by arrowheads) is shown in a, e, and i; area 30 in b, f, and j; area 23 in c, g, and k; and area 31 in d, h, and l. IG = induseum griseum, S = subiculum, arabic numbers = layers of the periarthocortical area 29, roman numbers = layers of the proiso- and isocortical areas 30, 23, and 31. Asterisks indicate the sulcus of the corpus callosum.

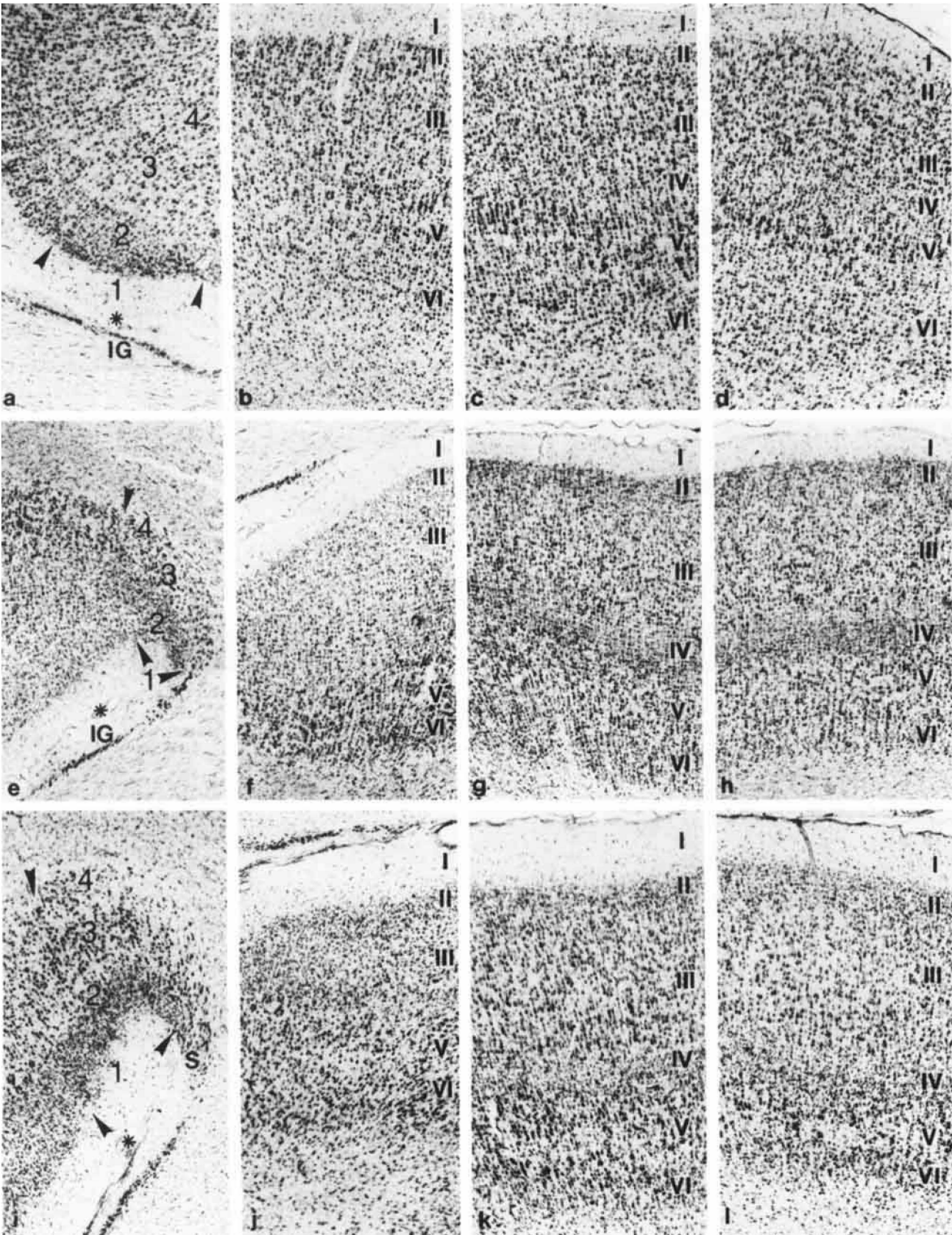


Figure 2

The border between areas 29 and 30 is defined by the presence of densely packed and very small cells in the outer pyramidal layer of area 29. Five laminae are observed in area 30 (granular retrosplenial area). Some primates, such as the rhesus monkey, have a clearly defined external granular layer (lamina II) (Fig. 2j), while in other species, such as *Saimiri sciureus* and *Lemur catta*, this layer is difficult to separate from the external pyramidal layer (Fig. 2b,f). Between laminae III and V some small granular cells are visible, indicating the beginnings of an internal granular layer, but these cells are not sufficiently clear to delimit a lamina IV for a quantitative approach. Therefore, the term agranular (Brodman, '09) or dysgranular (Sanides, '72; Vogt, '76; Braak, '79) is a good description of area 30's internal granular layer. This cytoarchitectural feature appears to be a first step toward an isocortex and this cortex can, therefore, be classified as proisocortex.

Area 30 can be subdivided into two subareas 30a and 30b in strepsirhines (prosimians excluding *Tarsius*), but not in haplorhines (*Tarsius* and anthropoids). This subdivision is based on differences in the packing density of the deeper layers (lower in area 30a, higher in area 30b) and laminae II and III (higher in area 30a, lower in area 30b). Since this subdivision is not found in haplorhines, the quantitative analysis is performed on area 30 as a unit.

Areas 23 and 31 show the typical six-layered isocortical structure (Fig. 2c,d, g,h, k,l) with a clearly discernable inner

granular layer (lamina IV) in anthropoids. Layer IV can also be detected in prosimians, but an intermingling of medium-sized pyramidal cells with the smaller granular cells produces a less-obvious layer compared with those of anthropoids. The presence of this distinctive inner granular layer separates area 23 from area 30 whereas the border between areas 23 and 31 is characterized by an increased thickness of lamina IV in area 31. Concomitantly, layer IV of area 31 has an increased number of granular cells and a decreased number of pyramidal cells. This feature is more obvious in anthropoids than in prosimians. In many cases, where the delineation by these cytoarchitectural criteria is difficult, the GLI plots, by showing a decrease in GLI of layers II and III in area 31 compared with the same layers in area 23, are especially helpful (Fig. 3).

The border between area 31 and the parietal or occipital isocortex can be defined by the appearance, in the latter isocortex, of very large pyramidal cells in layer V, bigger pyramidal cells in layer III, and a thinner layer IV that contains fewer granule cells. These features are found in all anthropoids. In prosimians only an increase in the size of the layer V pyramidal cells consistently separates area 31 from the parietal isocortex.

The areas 23 and 31 can be further subdivided in haplorhines but not in strepsirhines. The haplorhine subarea 23a has a thin layer IV consisting of small granular and larger pyramidal cells and a subarea 23b with a broader layer IV

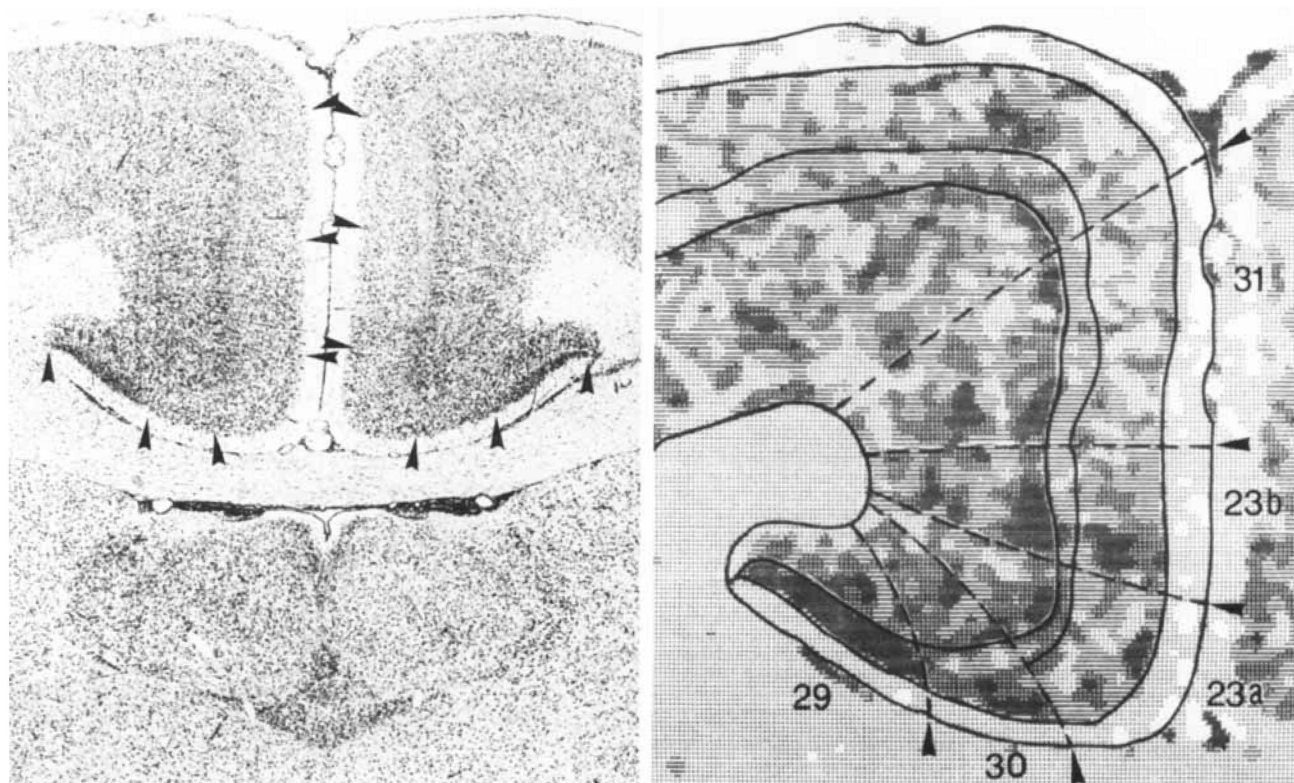


Fig. 3. Nissl-stained coronal section (20 μ m) (a) and respective computer plot (b) showing the gray-level index (GLI) distribution of the posterior

cingulate cortex in *Tarsius syrichta*. The areal borders are indicated by arrowheads.

that is almost free of pyramidal cell bodies. Subdivisions of area 31 are also observed in *Homo*, but the high interindividual cytoarchitectural variability combined with the distortions of lamination caused by the cingulate sulcus have to date precluded finding consistent criteria for subdividing this human cortex.

The rostral border between the posterior and anterior cingulate areas can be easily defined for area 29, because its outer, very densely packed pyramidal layer disappears here. A comparably sharp borderline has not yet been defined for the other posterior cingulate areas. The presently available methods, including GLI measurements, cannot adequately distinguish between the two regions because the cytoarchitecture changes slowly, producing a very broad transitional zone, especially in anthropoids. The following quantitative analyses are, therefore, based on sections from comparable positions immediately in front of the splenium of the corpus callosum.

Measurements

The volumetric proportions of the different cortical regions and layers are given in Table 1. A comparative analysis of the four cortical regions shows that the outer main lamina increases its size proportionately to the inner main lamina when the proisocortical area 30 is compared with allocortical area 29 and when the isocortical areas 23–31 are compared with area 30. The relative increase in areal proportion is found whether or not the granular layer is included as part of the outer main lamina (Fig. 4). The result was statistically significant among anthropoids ($t=5.7$, d.f. = 11, $P<.001$) and its subgroups the catarrhines ($t=13.2$, d.f. = 6, $P<.001$) and platyrrhines ($t=6.2$, d.f. = 4; $P<.01$) when each was studied separately. The outer main lamina of prosimian cortices, on the other hand, does not significantly increase in proportion from allo- to isocortex ($t=1.62$, d.f. = 4, n.s.).

A granular layer was delineated in areas 23 and 31. The granular layer in area 31 is relatively larger than it is in area 23 in all primates (Wilcoxon signed-ranks test $T_s=4$, $n=17$; $P<.005$). Both prosimians and anthropoids have a

statistically significant increase in the volumetric proportion of layer IV of area 31 compared to 23 (prosimian $T_s=0$, $n=5$, $P<.05$; anthropoid $T_s=4$, $n=12$, $P<.005$).

We found a significant decrease in the areal proportion of the molecular layer compared to the rest of the cortex going from allo- to isocortex. This was observed in each of the primate groups; anthropoids ($t=6.9$, d.f. = 11, $P<.001$), in its subgroups the catarrhines ($t=5.7$, d.f. = 6, $P<.01$) and platyrrhines ($t=5.8$, d.f. = 4; $P<.01$), and in prosimians ($t=7.6$, d.f. = 4, $P<.01$) (Fig. 5).

The mean GLI values vary between the different layers within an area and between cortical areas (Table 2). The molecular layer has an extremely low GLI, giving the standardized GLI values a high relative error. Consequently, further analysis of this layer was not performed. Compared to the inner main laminae, the outer main laminae decrease in GLIs from the allo- to isocortical areas. This means that the outer main laminae of isocortex have more space not occupied by cell somata than do the allocortical laminae. The reverse is true for the inner main laminae. Statistically significant changes in the mean GLI values were observed in each of the primate groups: anthropoids ($t=5.5$, d.f. = 11, $P<.001$), catarrhines ($t=5.4$, d.f. = 6, $P<.01$), platyrrhines ($t=2.9$, d.f. = 4, $P<.05$), and prosimians ($t=8.7$, d.f. = 4, $P<.001$) (Fig. 6).

In 11 out of the 17 primate species, the granular layer increases its GLI when area 31 is compared with area 23. This increase is not statistically significant for either the prosimians or anthropoids separately, but it is significant for all primates taken together (Wilcoxon signed-ranks test: $T_s=29$; $n=17$; $P<.025$).

DISCUSSION

The present investigation has found a parcellation of the posterior cingulate gyrus into four comparable areas in all primates. The borders between the periarchicortical area 29 (granular retrosplenial cortex) and the proisocortical area 30 (agranular retrosplenial cortex) and between the latter area and isocortical area 23 were easy to recognize. The boundary between areas 23 and 31 is more difficult to

TABLE 1. Volumetric Proportions of the Different Areas and Layers in the Posterior Cingulate Gyrus of 17 Primate Species¹

Species	Area 29			Area 30			Area 23				Area 31			
	M	O	I	M	O	I	M	O	G	I	M	O	G	I
<i>Lemur catta</i>	21	26	53	14	29	57	10	33	7	50	10	33	10	47
<i>Loris tardigradus</i>	29	34	37	22	34	44	13	31	9	47	14	34	12	40
<i>Perodicticus potto</i>	24	40	36	16	32	52	14	37	8	41	14	37	9	40
<i>Nycticebus coucang</i>	23	26	51	17	37	46	20	37	10	33	13	34	11	42
<i>Tarsius syrichta</i>	26	27	47	27	36	37	20	37	10	33	18	34	11	37
<i>Callithrix jacchus</i>	18	37	45	17	49	34	13	41	13	33	11	43	14	32
<i>Aotus trivirgatus</i>	24	38	38	19	42	39	11	38	12	39	11	46	12	31
<i>Alouatta seniculus</i>	19	41	40	20	43	37	19	42	11	28	14	45	13	29
<i>Saimiri sciureus</i>	19	40	41	13	52	35	13	42	14	31	9	44	13	34
<i>Cebus albifrons</i>	21	42	37	13	45	42	14	45	9	32	11	45	13	31
<i>Cercopithecus mitis</i>	19	33	48	16	44	40	17	41	12	30	12	41	15	32
<i>Erythrocebus patas</i>	19	33	48	16	43	41	16	43	11	30	10	45	12	33
<i>Macaca mulatta</i>	17	28	55	19	44	37	17	40	11	32	12	43	15	30
<i>Cercocebus albigena</i>	17	31	53	12	51	37	16	45	10	29	15	45	13	27
<i>Papio sp.</i>	19	25	56	17	44	39	15	42	9	34	15	46	11	28
<i>Pan troglodytes</i>	24	37	39	18	42	40	15	38	13	34	18	43	12	27
<i>Homo sapiens</i>	21	27	52	13	42	45	15	43	10	32	12	45	13	30

¹Abbreviations: M=molecular layer, O=outer main lamina, I=inner main lamina, G=granular layer.

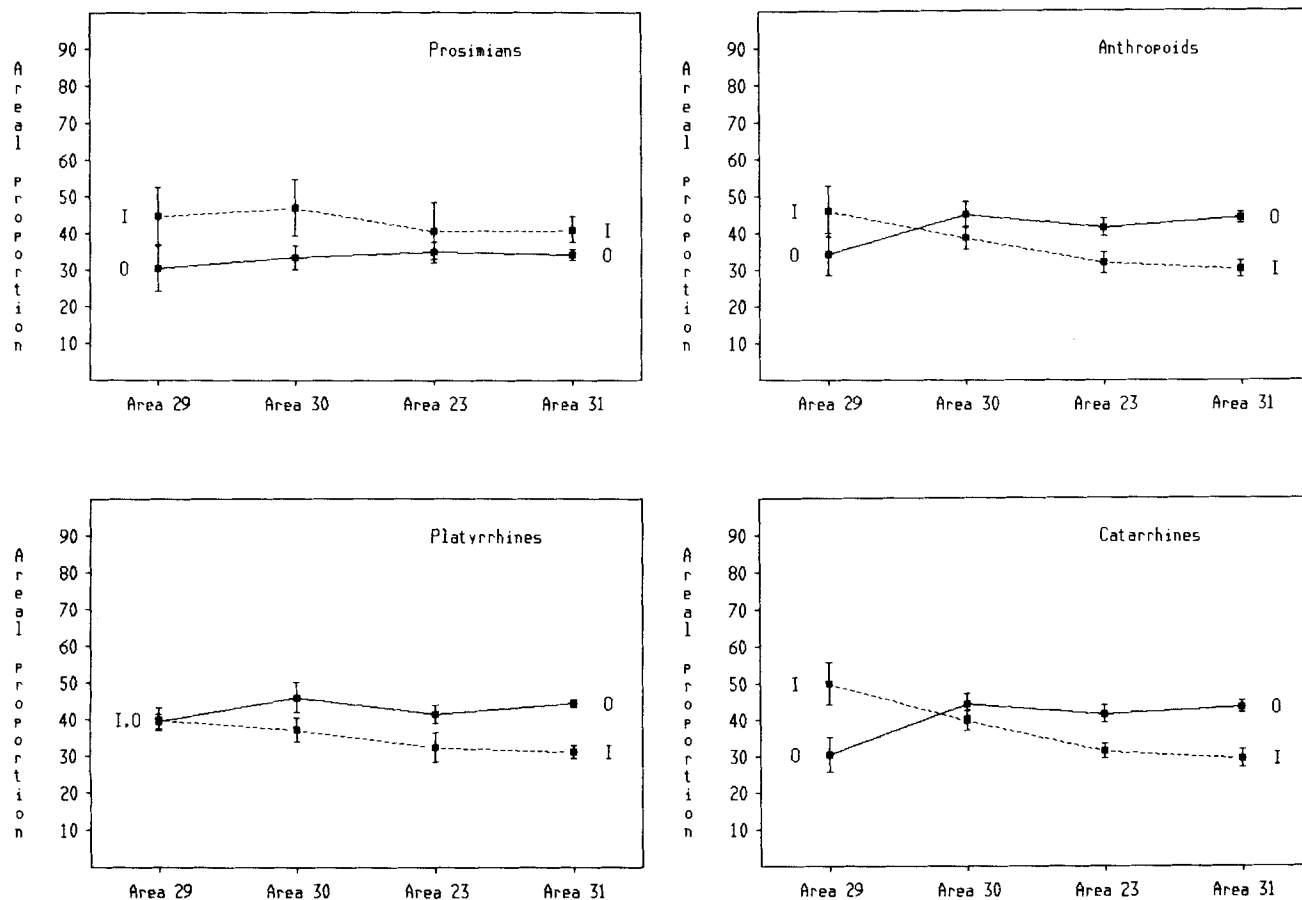


Fig. 4. Areal proportions of the outer (=O) and inner (=I) main laminae in different primate groups in the areas of the posterior cingulate cortex from allo- to isocortex. Mean values and standard deviations are given.

identify and was missed by Brodmann ('05, '08) in prosimians, but reported in humans (Brodmann, '09). The best-reproducible parameter for separating these areas is the increase in size and cell packing density in the granular layer of area 31. This is clearly shown by our measurements and furthermore demonstrates that when areas have a comparable basic cytoarchitecture, they can sometimes be differentiated by quantitative criteria.

Areas 23 and 31 are often not differentiated in connectional studies and are lumped together as area 23. The observations of Baleyrier and Mauguier ('85) on afferent connections to the posterior cingulate cortex from the medial pulvinar nucleus in *Papio* and *Macaca* show a decrease of label not only in transversing from area 23 to 30 but also in a dorsal direction within their "area 23." The decrease is not continuous, but shows a steplike form in the position where we have described the area 23/31 boundary. The distinction of these two areas by quantitative cytoarchitectonic methods, therefore, agrees with the results of connectional studies.

This comparative investigation of the allo- and isocortex in the posterior cingulate gyrus found differences in both

volumetric proportions of layers and GLI values. First, the primate allocortex has a relatively smaller outer main lamina and larger inner main and molecular laminae compared to the isocortical areas. Second, the outer main lamina of the allocortex has relatively higher GLI values than those found in the isocortex.

The volumetric proportion of layers was the parameter used to measure cortical field size because it is more reliable than thickness of cortical layers. In gyrencephalic brains, cortical thickness is strongly influenced by the geometric position of the measurements. For instance, the outer main lamina is thinner on a gyral convexity than it is in the depth of a sulcus (Bok, '36). The reverse is true for the inner main lamina. Consequently, a parameter that is independent of these geometric constraints is preferable. Given that the outer cortical layers at a convexity are thinner, but longer in a plane parallel to the section plane than are the inner layers, and that the reverse is true for the inner main stratum (i.e., that the length and depth of a cortical layer are inversely correlated), then a measurement of areal proportion throughout these transitions is more stable than a measurement of cortical thickness. Thus,

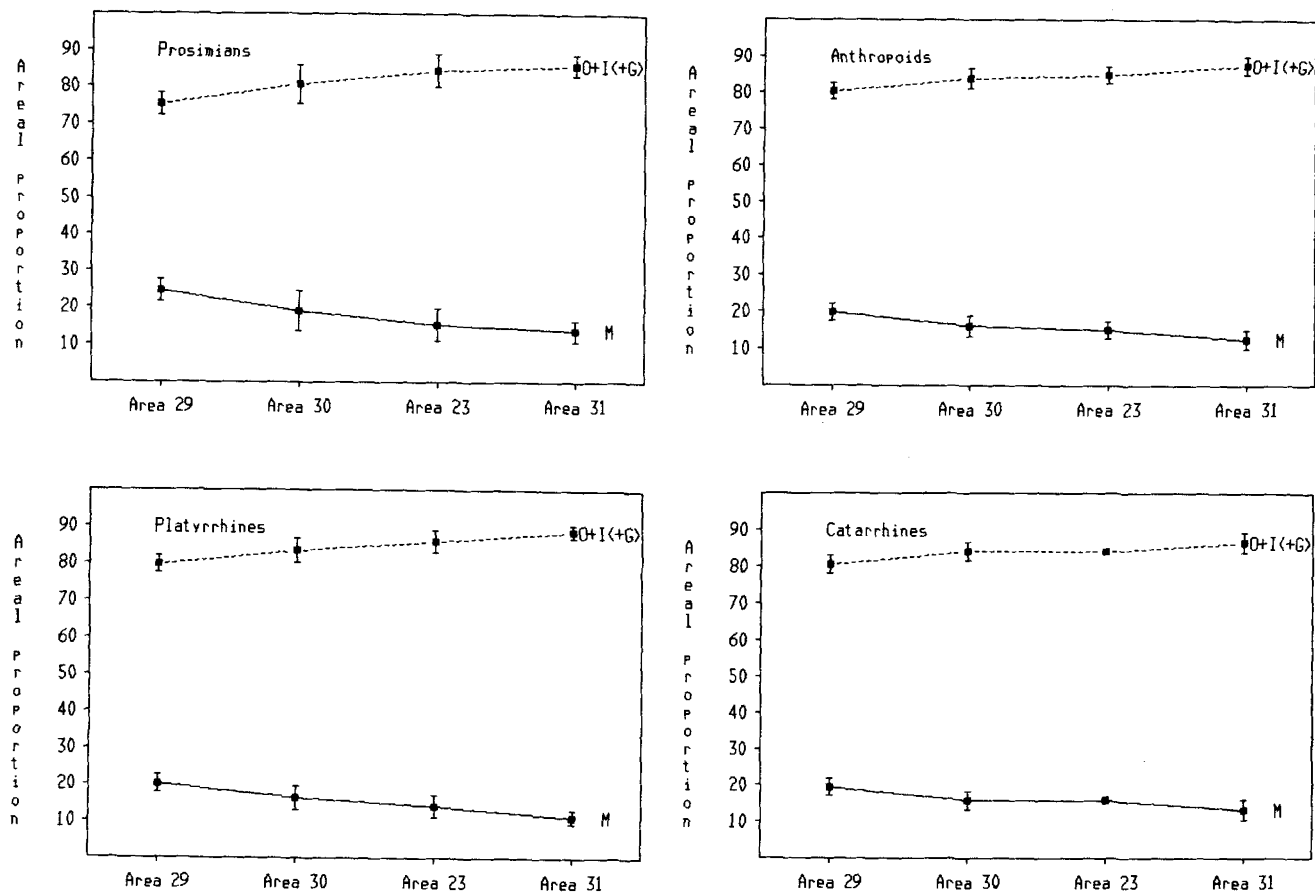


Fig. 5. Areal proportions of the molecular (=M) layer and outer plus granular plus inner main laminae [=O+I+(G)] in different primate groups in the areas of the posterior cingulate cortex from allo- to isocortex. Mean values and standard deviations are given.

TABLE 2. Grey Level Indices (GLI) of the Different Areas and Layers in the Posterior Cingulate Gyrus of 17 Primate Species¹

Species	Area 29		Area 30		Area 23			Area 31		
	O	I	O	I	O	G	I	O	G	I
<i>Lemur catta</i>	29.2	15.7	24.4	17.9	21.7	23.5	19.5	21.5	24.3	19.9
<i>Loris tardigradus</i>	28.4	22.3	24.1	23.5	26.9	28.3	27.9	24.8	29.6	29.4
<i>Perodicticus potto</i>	27.0	19.4	17.3	17.4	15.0	17.0	18.6	14.9	16.4	18.6
<i>Nycticebus coucang</i>	20.5	15.1	14.0	13.9	12.5	13.4	16.3	13.4	13.7	17.2
<i>Tarsius syrichta</i>	40.1	30.5	28.8	29.1	26.7	29.8	30.1	22.8	27.2	26.8
<i>Callithrix jacchus</i>	30.0	24.3	25.6	24.6	25.7	29.4	27.4	26.6	30.2	26.6
<i>Aotus trivirgatus</i>	27.5	21.0	17.9	22.9	16.2	25.3	21.9	12.5	24.0	18.8
<i>Alouatta seniculus</i>	14.1	16.9	12.7	13.8	9.6	9.3	11.4	8.5	10.9	13.7
<i>Saimiri sciureus</i>	34.4	25.4	26.5	34.3	25.5	39.8	37.7	25.7	42.1	32.9
<i>Cebus albifrons</i>	20.8	20.0	17.9	23.0	19.5	27.1	26.5	18.1	24.5	21.6
<i>Cercopithecus mitis</i>	24.4	18.2	22.8	20.0	22.9	27.1	22.6	23.6	30.2	24.0
<i>Erythrocebus patas</i>	18.9	13.7	16.5	15.5	11.6	20.2	17.3	14.3	25.4	21.6
<i>Macaca mulatta</i>	22.4	16.1	18.3	15.8	18.7	19.6	19.9	14.9	17.5	17.8
<i>Cercocebus albigena</i>	16.3	16.8	14.3	14.9	8.9	11.8	13.5	9.5	14.6	13.5
<i>Papio</i> sp.	17.9	14.0	14.3	12.9	12.2	14.9	15.0	11.1	15.6	15.6
<i>Pan troglodytes</i>	9.0	9.0	8.8	8.9	9.0	8.8	8.1	6.4	8.4	7.9
<i>Homo sapiens</i>	10.0	6.9	9.8	8.2	9.7	11.0	9.4	10.1	12.4	10.1

¹Abbreviations as in Table 1.

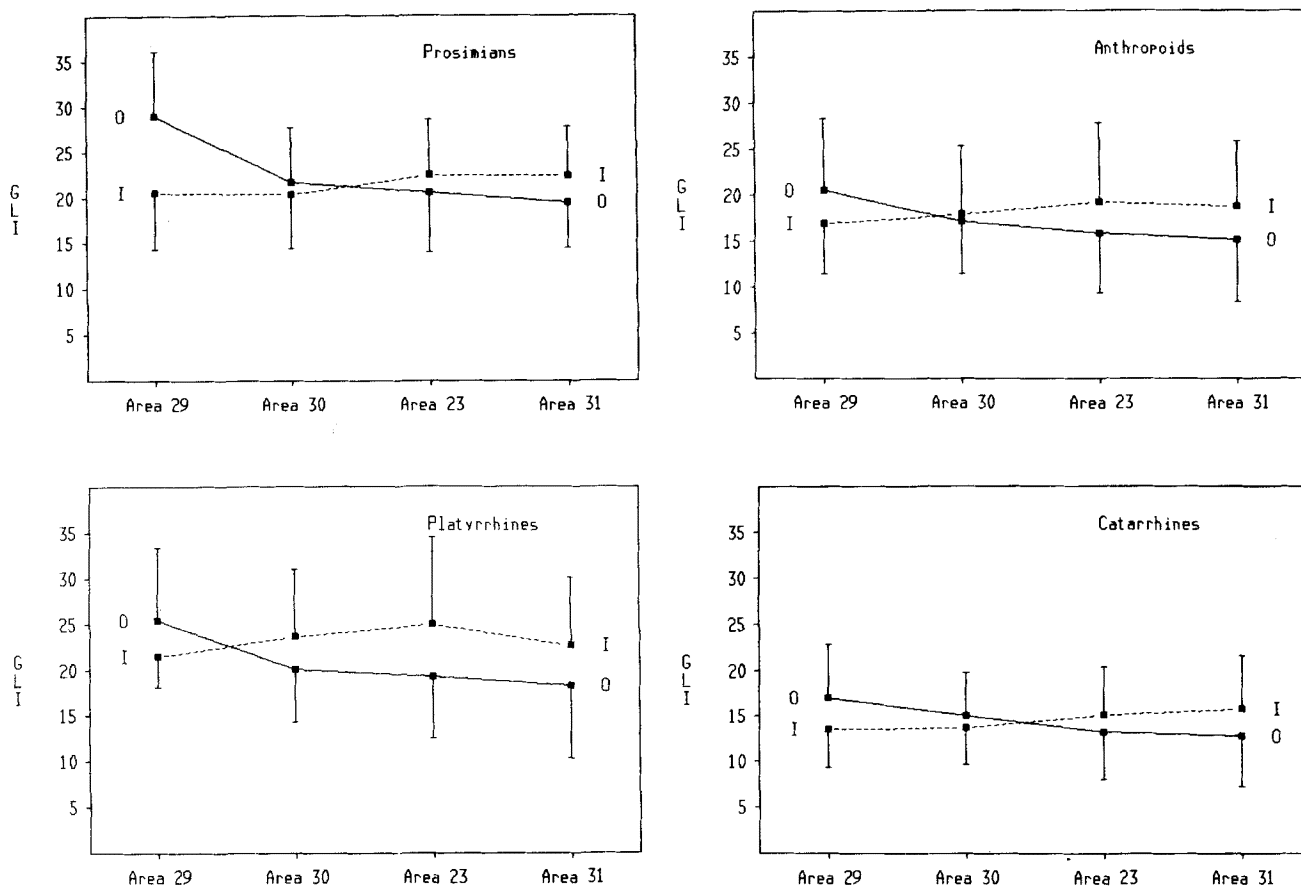


Fig. 6. GLI values of the outer (=O) and inner (=I) main laminae in different primate groups in the areas of the posterior cingulate cortex from allo- to isocortex. Mean values and standard deviations are given.

areal or volumetric proportions are more consistent morphometric features than depth.

Layer IV increases in volumetric proportion from area 23 to area 31. Corroborating Vogt's ('76) results in rhesus monkeys, we also found some small cells intermingled with pyramidal cells in layers III and V of area 30. The increase in areal proportion of layer IV in all primate groups is a quantitative proof for the structural specialization of a receptive cortical field in the isocortex.

The molecular layer decreases in volumetric proportion from allo- to isocortex in all primates. The functional meaning of this result is not clear, but more myelinated fibers are found in layer I of area 29 than in the proiso- or isocortex (Meynert, 1872; Stephan, '75).

The measure GLI permits a further refinement of the description of changes within the main laminae by separating and measuring the spaces occupied by neuronal and glial cell bodies from the neuropil. A high GLI value is associated with high cell body density and less space for neuropil. Differences in cortical cell densities are associated with differences in various structures like dendritic trees and axonal arborizations (Eayrs and Goodhead, '59; Vogt, '76). An increase in dendrites and/or axons indicates a capacity for more connections and is correlated with low GLI value. Since the density of glial cells does not vary system-

atically throughout the layers (Wree et al., '82), changes in GLI are thought to indicate differences in neuronal density.

We found GLI values to be lower in the outer main lamina and higher in the inner main lamina when the isocortex is compared with allocortex. This corroborates Vogt's ('76) impression that the outer main lamina of his granular retrosplenial cortex (our area 29) is more densely packed than the comparable lamina in his agranular retrosplenial cortex (our area 30). Isocortical areas 23 and 31 have an even greater decrease in cell density in the supragranular layers. Vogt ('76) explained the differences in cell density in areas 29 and 30 as the result of more elaborate dendritic fields of pyramidal cells in area 30. Presumably, the same explanation can be extended to areas 23 and 31.

In anthropoids, the granular layer GLI increases from area 23 to area 31. A comparable increase was not found in prosimians. Ramón y Cajal ('09) and Lorente de Nó ('33, '34) observed that more complex brains have an increasing density of neurons with short axons. Since layer IV of area 23 has many short axon cells (Vogt, '76), our measurements suggest that layer IV of area 31 has more granule cells than area 23. Combining the GLI findings with the increasing volumetric proportions of layer IV shows an even greater increase in the granular layer in area 31 compared to area 23 and this increase is found in all primates. An

increase in granule layer size, a cortical transitional feature called "granularization" by von Economo and Koskinas ('25), characterizes the shift from area 23 to area 31 in the primate brain.

Differences in areal proportions of layers and GLI values were found to covary, so that the outer main laminae of the isocortex combine bigger volumetric proportions with lower GLI values than found in the allocortex. Isocortex differs from allocortex by having relatively more neuropil space in its outer main lamina than in its inner, and in having this zone be relatively larger.

The cell layers of the outer main lamina are the major receptive and information processing zones of the cortex (Szentágothai, '83). The inner main lamina includes both incoming fibers that are passing through to the outer layers and most of the cortical output cells. While this is a rough outline of cortical structure, it suggests in a general way that an increase in the size of the outer layers with a concomitant decrease in the inner layers reflects a change toward more integration and information processing. Such a trend characterizes the shift from allo- to isocortex within the primate posterior cingulate region. In all primate species analyzed in this paper differences were statistically significant in anthropoids and in their infraorders, the platyrrhines and catarrhines. The differences between the volumetric proportions of the laminae were not statistically significant among the prosimians because the allo- and isocortical areas are more alike and the variability is bigger in this parameter than in the other primates. It is not clear at this point why the prosimians have a particularly variable posterior cingulate gyrus. The present results quantitatively corroborate an increasing complexity in laminar structure of isocortex such as has often been qualitatively described (Brodman, '09; Ramón y Cajal, '09; von Economo and Koskinas, '25; Rose, '27; Stephan, '75; Vogt, '76), a complexity that could support increasing cortical integrative capacities beyond that found in allocortex.

The present investigation not only corroborates the concept of the posterior cingulate region as a transitional cortex between allo- and isocortex (Loo, '31; Abbie, '40, '42; Filimonoff, '47; Stephan, '75; Vogt, '76), but has determined some of the actual transitional features and gradients in primates. The increase in number of layers from the periaarchicortical area 29 to the isocortical areas 23 and 31 confirms Vogt's ('76) description in the rhesus for a broad range of primates. Additionally, the first appearance of some granular cells between the outer and inner pyramidal layers in the proisocortical area 30 (Vogt, '76; present results) and their further differentiation into a typical inner granular layer in areas 23 and 31 clearly separate allocortex (area 29) from isocortex (areas 30, 23, 31). The increase in volumetric proportion and packing density of neurons ("granularization") in layer IV forms a gradient (Sanides, '62) in cortical differentiation that we have shown to be a quantitatively significant phenomenon. Since neurons in the primate isocortical layer IV are mostly interneurons, their significant increase from allo- to neocortex can be interpreted as the latter cortex having a structural basis for increased processing of afferent information. The decrease of GLI values in the outer main lamina indicates an increasing neuropil content in these layers and suggests an enhancement of integrative properties. The quantitative analysis of the allo- to isocortical transition can be summarized, therefore, as a progressive structural differentiation

permitting an increasing amount of intracortical information-processing capacities.

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